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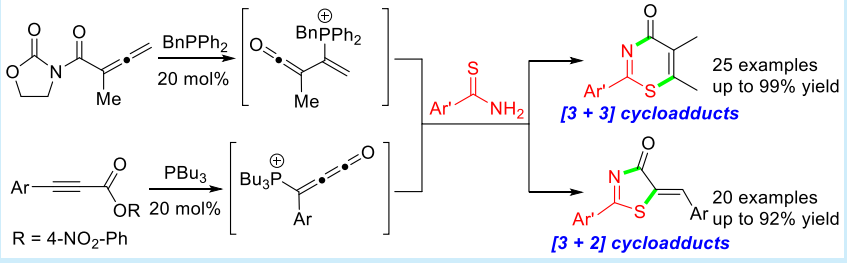
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Phosphine-Catalyzed Annulations Based on [3+3] and [3+2] Trapping of Ketene Intermediates with Thioamides

Yu-Hao Wang,[†] Zhen-Ni Zhao,[†] Subarna Jyoti Kalita, and Yi-Yong Huang*

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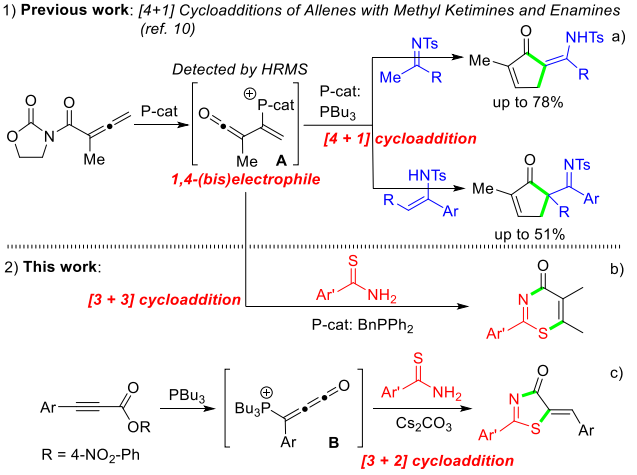
ABSTRACT: With the aim of developing novel annulations via ketene intermediates, allenyl imide and alkynoates bearing good leaving groups are used for their function in a tandem conjugate addition−elimination reaction (SN2′ type) promoted by nucleophilic phosphine catalysts. By utilizing thioamides as 1*S*,3*N*-bis-nucleophiles, [3+3] and [3+2] annulations have been established to allow rapid access to 1,3-thiazin-4-ones and 5-alkenyl thiazolones in high yields, respectively. Furthermore, the possible reaction mechanisms are proposed on the basis of deuterium labeling experiments and density functional theory calculations.

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Ever since the groundbreaking report on elucidating the structure of diphenylketene in 1905 by Staudinger,[1] the generation and synthetic utility of ketenes have attracted a great deal of interest from organic chemists.[2] As a class of cumulenes, ketenes possess highly reactive carbonyl and alkenyl groups, which makes them very useful chemical entities and versatile reactive intermediates. As demonstrated in industrial applications[3] and academic research, including polymer science[4] and organic synthesis such as total syntheses[5] and methodology development,[6] ketenes play a vital role. In particular, they are effective in catalytic cycloaddition for various types of carbo- and heterocycle synthesis.[7] Therefore, the discovery of novel ketene intermediates and ketene-based reactions represents a significant task.

By far, the most commonly used approaches for generating ketenes are base-mediated elimination of HCl from acyl chlorides[8] and Wolff rearrangement of diazo ketones.[9] In our previous work, a phosphine catalyst promoted the generation of reactive vinylphosphonium ketene species **A** through 1,4-addition of PBu_3 and dissociation of a 2-oxazolidinyl anion from allenyl imide, which was captured by various 1*C*,1*C*-bis-nucleophiles through [4+1] cycloaddition for the bis-electrophilicity (Scheme 1a).[10] On the basis of this work, we envisaged that (1) thioamides[11] can be used as potential 1*S*,3*N*-bis-nucleophiles to capture vinylphosphonium-ketene intermediate **A** to install N,S-containing heterocycles (Scheme 1b) and (2) the employment of alkynoates containing a good leaving group[12] may provide a [3]cumulene type of

Scheme 1. Several Annulations Involving Ketene Intermediates



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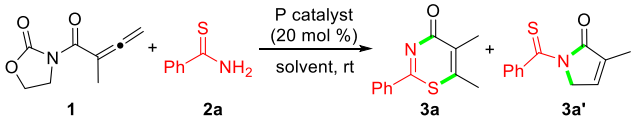
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phosphonium-ketene intermediate **B** through nucleophilic phosphine catalysis, which may display unique reactivity in the novel cycloaddition of thioamides (Scheme 1c). With the purpose of advancing the synthetic potential of the phosphonium-ketene intermediates and developing new types of reactions, we perform these studies aiming at two types of annulations with thioamides for building N- and S-containing heterocycles, viz., 1,3-thiazin-4-one and 5-alkenyl thiazolones.

1,3-Thiazin-4-ones make up an important family of six-membered heterocycles because of their bioactive activities.[13] For instance, in Figure S1, *BTZ043* can kill *Mycobacterium tuberculosis*.[14] The natural product Thiasporine A, isolated from the marine-derived *Actinomycetospora chlora* SNC-032, showed cytotoxicity against the non-small-cell lung cancer cell line H2122.[15] Thus, the development of an efficient approach to generating 1,3-thiazin-4-one core has attracted a great deal of interest.[16] Nucleophilic phosphine catalysis has emerged as a powerful tool for the installation of various heterocycles, relying on the interception of the reactive intermediate, via the 1,4-addition of phosphine to activated allene, by electrophiles or nucleophiles.[17] To the best of our knowledge, there is no precedent for using thioamides as bis-nucleophiles in the allene-based cycloadditions.[18] Nevertheless, the formidable challenge of controlling the chemo- and regioselectivities resulting from the multiple reaction sites in the substrates will be addressed.

To test the feasibility of installing a 1,3-thiazin-4-one derivative from allenyl imide **1** and thiobenzamide **2a**, PBu_3 was first studied in dichloroethane at room temperature (rt) (Table 1, entry 1). 4*H*-1,3-Thiazin-4-one **3a** (49%) and 2*H*-

Table 1. Optimized Conditions Using Allenyl Imide[a]



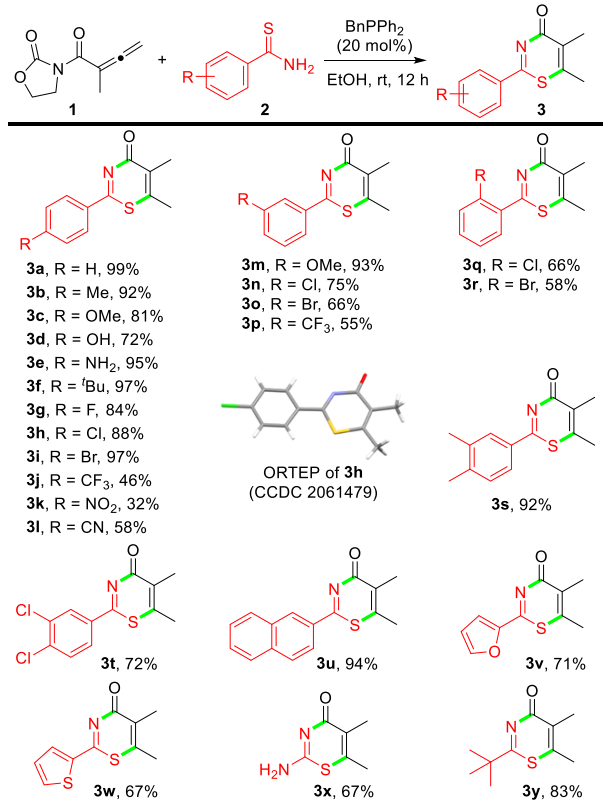
entry	P catalyst	solvent	time (h)	yield (%)[b] 3a	yield (%)[b] 3a′
1	PBu_3	$ClCH_2CH_2Cl$	1	49	24
2	$PPhMe_2$	$ClCH_2CH_2Cl$	1	50	21
3	PPh_2Me	$ClCH_2CH_2Cl$	1	71	23
4	PPh_3	$ClCH_2CH_2Cl$	2	78	16
5	$BnPPh_2$	$ClCH_2CH_2Cl$	1	82	trace
6	$BnPPh_2$	$ClCH_2CH_2Cl$	12	89	trace
7	–	$ClCH_2CH_2Cl$	12	complex mixture	
8	$BnPPh_2$	CH_2Cl_2	12	91	trace
9	$BnPPh_2$	THF	12	complex mixture	
10	$BnPPh_2$	toluene	12	77	trace
11	$BnPPh_2$	toluene	12	64	trace
12	$BnPPh_2$	EtOH	12	99	–

[a]**1** (0.10 mmol), **2** (0.12 mmol), P catalyst (20 mol %), and solvent (1.0 mL). [b]Isolated yield.

pyrrol-2-one **3a′** (24%) generated through respective [3+3] and [4+1] cycloadditions were isolated. The structure of **3a** was assigned by NMR and HRMS and further confirmed by X-ray crystallography,[19] and thus, the hypothesis of using thioamide as a 1*S*,3*N*-bis-nucleophile to trap the proposed vinylphosphonium ketene intermediate was validated. The use of less nucleophilic phosphines, such as $PhPMe_2$, Ph_2PMe, and PPh_3, could increase the yield of product **3a** gradually, whereas a small amount of adduct **3a′** was still isolated (entries 2−4). To our delight, in the presence of a $BnPPh_2$ catalyst, the yield of **3a** reached 82% with marked suppression of the formation of **3a′** (entry 5). When the reaction time was increased to 12 h, the yield was further improved to 89% (entry 6). The comparison without any catalysts indicated a messy reaction mixture (entry 7). In addition, the screening of other solvents, including CH_2Cl_2, THF, toluene, benzene, and ethanol, revealed that the protic solvent was preferred, because the yield of **3a** was dramatically increased to 99% in ethanol (entry 12).

With the optimized reaction conditions in hand, we focused on the scope of various thioamides (Scheme 2). Gratifyingly, a

Scheme 2. Scope of Thioamides with Allene 1[a]



[a]**1** (0.10 mmol), **2** (0.12 mmol), $BnPPh_2$ (20 mol %), and EtOH (1.0 mL). Isolated yields.

range of 4*H*-1,3-thiazin-4-one derivatives **3a−y** were synthesized in 32−99% yields. First, the effect of substituents at the *para* position of thioarylamides on the yield was studied. When electron-donating groups were introduced, the Me group provided a yield (92%) higher than that of the OMe group (81%). Substrates having an active hydrogen group were tolerated in the reaction, and the performance of the NH_2 group (95%) was much better than that of the OH group (72%). Product **3f** with a ^{t}Bu group was isolated in a yield comparable to that of **3a**. The use of substrates bearing halogen substituents (F, Cl, and Br) provided the corresponding products **3g−i**, respectively, in 84−97% yields. Thereafter, the substrates with electron-withdrawing groups (CF_3, NO_2, and CN) were applied to afford <60% yields (**3j−l**), which resulted from the undesired aza-Michael addition of thioamide to allene. Next, diverse substituents at the *meta* position of

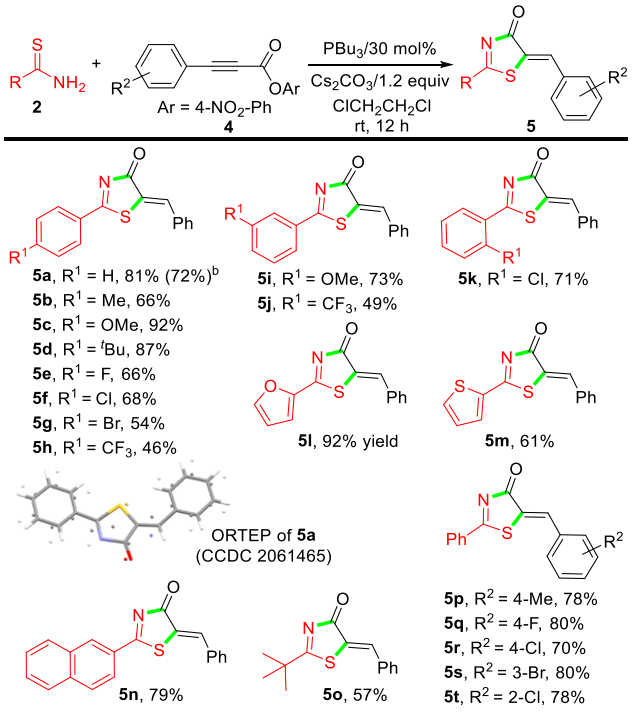
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thioarylamides were investigated. Compared with the same group at the *para* position, the OMe and CF_3 groups delivered superior outcome (**3m**, 93%; **3p**, 55%), albeit we produced much lower yields for the Cl and Br groups (**3n**, 75%; **3o**, 66%). When using *ortho*-substituted substrates (Cl and Br), moderate yields were obtained (**3q**, 66%; **3r**, 58%). The employment of *meta*,*para*-disubstituted thioarylamides indicated that Me groups (**3s**, 92%) were more suitable than Cl groups (**3t**, 72%). The synthesis of the 2-naphthyl derivative of 4*H*-1,3-thiazin-4-one (**3u**) was achieved in 94% yield. Likewise, heterocyclic substrates bearing 2-furyl and 2-thienyl groups underwent [3+3] cycloaddition to give products **3v** and **3w** in 71% and 67% yields, respectively. In addition, 4*H*-1,3-thiazin-4-ones **3x** and **3y** from thiourea and thiopivalamide were afforded in 67% and 83% yields, respectively.

Using the tandem 1,4-addition of phosphine and elimination of a leaving group (SN2′ type reaction) described above to generate ketene intermediates, we continued to pursue new candidates to check the reactivity with thioamides. 3-Arylpropiolates bearing 4-nitrophenyl as a good leaving group can act as Michael acceptors and have been successfully utilized in annulations involving alkyne acyl ammonium and azolium intermediates through nucleophilic amine and N-heterocyclic carbene (NHC) catalysis.[20] Herein, 4-nitrophenyl 3-arylpropiolates were studied for the first time in nucleophilic phosphine-catalyzed annulations, and the key intermediate involved was quite unlike those in amine and NHC catalysis for the 1,4-addition model instead of the 1,2-addition model of the catalyst to arylpropiolate. The optimization results using thiobenzamide **2a** and 4-nitrophenyl 3-phenylpropiolate **4a** were shown in Table S1. Eventually, [3+2] cycloaddition through C(O)−N and α-S−C bond formation occurred, delivering 5-alkenyl thiazolone **5a** in ≤81% yield in the presence of 30 mol % PBu_3 and Cs_2CO_3 (1.2 equiv) (Scheme 3). Herein, the role of Cs_2CO_3 is to neutralize the 4-nitrophenol byproduct ($pK_a \sim 7.22$), which is more acidic than that of the 2-oxazolidinone byproduct mentioned above ($pK_a \sim 12.78$). It should be noted that 5-alkenyl thiazolones have been recognized as important moieties in many bioactive molecules (Figure S1)[21] and as substrates for synthetic methodology development.[22] Typically, 5-alkenyl thiazolones are synthesized by Knoevenagel condensation of thiazolin-4-ones and aldehydes.[23] It should be mentioned that 4 Å molecular sieves (MS) were detrimental to the yield.

The substituent aryl groups for two substrates were validated under the optimized reaction conditions, and the results are summarized in Scheme 3. By fixing the use of alkynoate **4a**, we investigated the scope of various aromatic, heteroaromatic, and aliphatic thioamides (≤92% yield). Thioarylamides with a Me group at the *para* position provided yields comparable to or slightly higher than those of halogen groups (54−68%), while much higher yields were obtained when incorporating OMe and ^{t}Bu groups (**5c**, 92%; **5d**, 87%). The electron-withdrawing group (CF_3) at the same position resulted in a lower yield (**5h**, 46%). The same tendency of the substituent effect was also observed at the *meta* position, and a better result was achieved for the electron-donating group (**5i**, OMe, 73%; **5j**, CF_3, 49%). When a Cl group was introduced at the *ortho* position, cycloadduct **5k** was formed in 71% yield. Notably, the performance of the substrate with a 2-furyl group was very superior to that of the 2-thienyl case (**5l**, 92%; **5m**, 61%). Moreover, 2-naphthyl-substituted thioamide was examined to produce **5n** in 79% yield. *tert*-Butyl thioamide was also

Scheme 3. Scope of Thioamides and 3-Arylpropiolates[a]



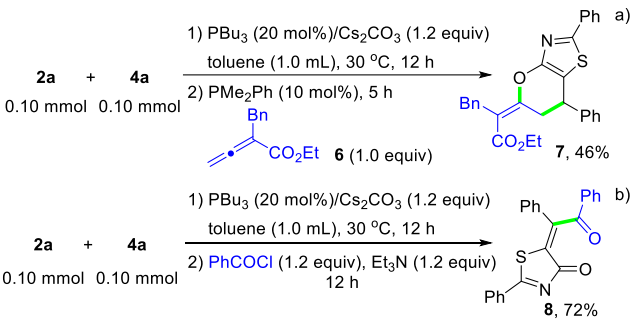
[a]**2** (0.12 mmol), **4** (0.10 mmol), Cs_2CO_3 (0.12 mmol), PBu_3 (30 mol %), and $ClCH_2CH_2Cl$ (1.0 mL). Isolated yields based on **4**. [b]With 1.0 mmol of **4**.

tolerated, and the corresponding product **5o** was afforded in 57% yield. Next, other five 3-arylpropiolates underwent the [3+2] cycloaddition of thiobenzamide **2a** to give the corresponding 5-alkenyl thiazolones **5p**−**t** in 70−80% yields.

5-Alkenyl thiazolones feature the α,β-unsaturated carbonyl scaffold. To demonstrate the synthetic versatility of 5-alkenyl thiazolones empowered by nucleophilic phosphine catalysis, two sets of one-pot sequential reactions were examined. Under the standard reaction conditions mentioned above, thiazolone **5a** was first generated, which then underwent Me_2PPh-catalyzed [4+2] annulation of α-Bn-substituted allenoate **6** for 6,7-dihydro-5*H*-pyrano[2,3-*d*]thiazole **7** (Scheme 4a),[24] and PBu_3-catalyzed formal β-acylation with acyl chloride for 5-arylidene thiazolone **8** (Scheme 4b).[25]

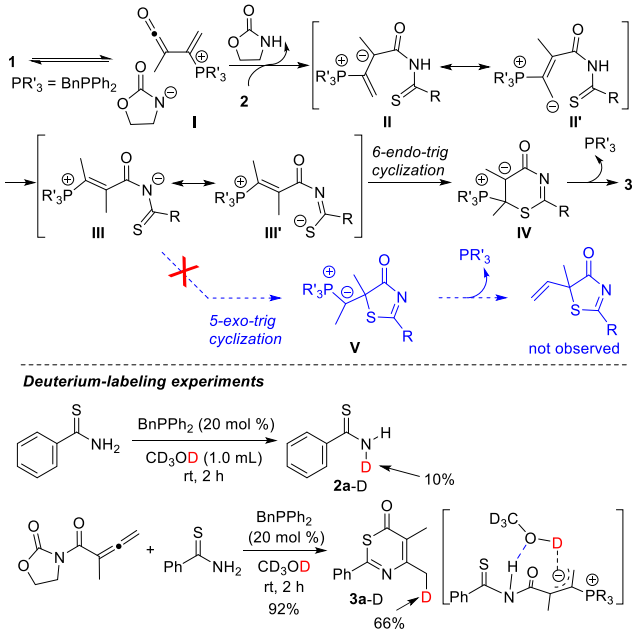
The proposed reaction mechanism for the syntheses of 1,3-thiazin-4-one **3** begins with formation of vinylphosphonium ketene intermediate **I** from **1** and $BnPPh_2$ (Scheme 5). The amide nitrogen anion, generated via the deprotonation of **2** by the oxazolidinyl anion, adds preferentially to the sp-hybridized

Scheme 4. Synthetic Utilities of 5-Alkenyl Thiazolone



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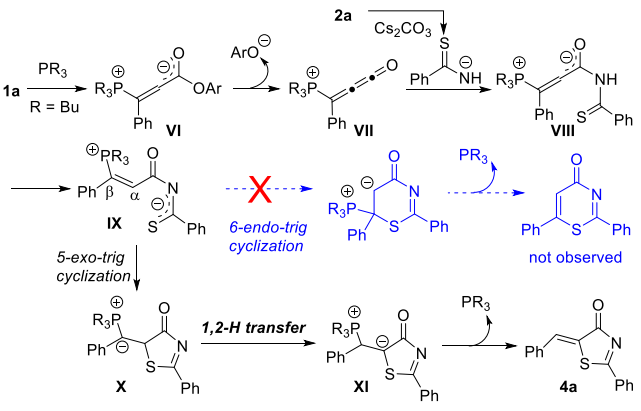
Scheme 5. Proposed Reaction Pathway for 1,3-Thiazin-4-one 3 and Deuterium Labeling Experiments



carbon of ketene intermediate **I** to give imide intermediate **II**. Thereafter, anion exchange through 1,5-H transfer occurs to generate nitrogen anion **III**, which undergoes *6-endo-trig* cyclization by way of sulfur anion resonance form **III′** followed by the elimination of $BnPPh_2$ to give 1,3-thiazin-4-one **3**. To understand the key 1,5-H transfer step (**II** → **III**), the deuterium labeling experiments were carried out in the CD_3OD solvent. Because only 10% deuterium was incorporated into the thioamide (**2a**-D) in the absence of allenyl imide **1**, the 66% deuterium incorporated into the methyl group of product **3a**-D may be mainly attributed to CD_3OD participating in the 1,5-H transfer step instead from **2a**-D, which was also consistent with the dramatic effect on the yield in an alcohol solvent.

Compared with the [3+3] cycloaddition of allenyl imide and thioamide, the proposed reaction mechanism of [3+2] cycloaddition using alkynoate also comprises a tandem phospha-Michael addition/elimination/nucleophilic addition of ketene intermediate by the N anion sequence at the beginning (Scheme 6), forming similar types of intermediates

Scheme 6. Proposed Reaction Pathway for 5-Alkenyl Thiazolones



with different substituents in the C=C bond region (**III** and **IX**). Thereafter, two distinct models of cyclization were observed. For the former case, **V** could likely be formed but without a good termination pathway, and it would go backward to **III**. On the contrary, starting from alkynoate and thioamide, the intramolecular addition[26] (*5-exo-trig* cyclization) of intermediate **IX** to provide [3+2] cycloadduct **4a** is preferred rather than *6-endo-trig* cyclization probably due to the more steric hindrance at the *β*-position of intermediate **IX** than that at the *α*-position.

Furthermore, the transformation from intermediate **X** to **XI** involving 1,2-H transfer was understood through density functional theory (DFT) calculations by using the Gaussian 16 suite of programs (see the details in the Supporting Information). Four possible approaches were considered (see Figure S2). Direct 1,2-H transfer without any assistance was first excluded because of a high energy barrier ($\Delta G^{\ddagger}$ = 38.8 kcal/mol). H_2O-assisted 1,2-H transfer in a stepwise manner appears to be more favorable, which exhibits activation energies ($\Delta G^{\ddagger}$ = 18.0 and 8.4 kcal/mol) much lower than that of the synchronized manner ($\Delta G^{\ddagger}$ = 28.1 kcal/mol); it is consistent with the inferior result upon using 4 Å MS to remove a trace amount of water (see the result in entry 12 of Table S1).[27] By comparison, the 1,2-H transfer between two molecules of intermediate **X** was also studied, which seems less possible on the basis of the Gibbs free energy change profiles (see the details in the Supporting Information).

In summary, we have demonstrated the novel phosphine-catalyzed [3+3] and [3+2] annulations of allenyl imide and alkynoates with thioamides involving vinylphosphonium and [3]cumulene types of phosphonium ketene intermediates, respectively, providing new entries to 1,3-thiazin-4-ones (25 examples, ≤99% yield) and 5-alkenyl thiazolones (20 examples, ≤92% yield) under mild conditions. The key proton transfer steps were understood on the basis of deuterium labeling experiments and DFT calculations. The synthetic utilities of 5-alkenyl thiazolone were shown through two sets of sequential phosphine-catalyzed derivatizations in one pot. The methods reported here harness the power of ketene intermediates and will open up the opportunity to enrich the nucleophilic phosphine catalysis strategies for creating new types of heterocycles.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02803.

Experimental procedures, characterization data, and copies of ^{1}H, ^{13}C, and ^{19}F NMR spectra of all new compounds (PDF)

Accession Codes

CCDC 2061465 and 2061479 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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